

Iron Triangle

A Path-Elected, Quorum-Signed Clearing Exchange for the MultiPlanetary System

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Abstract

Decision in one sentence. Keep each settlement’s local book durable and private, let any of three anchor replicas at Earth, Mercury, and Venus serve as the initiator for a cross-settlement transaction, chosen per trade by the originating cell for the best round-trip path to both parties, and require a two-of-three quorum signature before any commit is final. Local cells lock assets before any remote instruction; a signed quorum COMMIT is the only authority; a packet receipt, timeout, or stale screen never moves an asset. The Triad is useful when connected and safe when not.

1 Commission and Scope

The commission asks for a working MultiPlanetary Exchange System: markets, clearing, settlement, and supporting institutions that keep operating while the planets and relays move. The story is set on 2126-09-22, but every body starts where the real solar system is on 2026-09-22. The epoch is $t = 0$; all times are elapsed model hours unless stated.

Design principles.

1. **Conserve first.** No asset supports two uses at once. Every guarantee, loan, payout, and unit of collateral comes from the opening balance sheet through an explicit, valid transaction.
2. **Name the condition.** Every guarantee states when it holds and what participants pay or give up.
3. **Probability is part of the promise.** No conditional time counts as certain; every direct-service time carries its success probability and fallback.
4. **Fail into a funded owner.** No obligation is cleared automatically. Unresolved obligations keep a named, funded owner.

Implemented vs extension. Implemented: local books and locks, signed two-phase transfers, a quorum-capable registry with per-trade path-elected initiator, Ares spot, fully escrowed fixed loans, capped cash-settled futures, capped options, simple bonds, a fully backed settlement unit, versioned price observations, and recovery. Extension only: malicious or colluding operators, a third relay, higher-fidelity gravity, relativistic timekeeping, and operational key compromise.

2 Design Overview

The Triad is a cooperating exchange with three anchors: Earth, Mercury, and Venus. Each anchor runs a replica of the authoritative ledger and can act as the initiator for a cross-settlement transaction. The other six settlements — Mars, Ceres, Jupiter, Saturn, Uranus, Neptune — participate as client locations through local gateway agents that hold local books and local locks.

The choices that follow from the brief are:

- **Anchors:** Earth, Mercury, Venus. The lowest-loss inner-system triangle in the supplied 200-year scan.
- **Authority:** any of the three anchors can serve as the initiator for a transaction, chosen per trade by the originating cell for best round-trip path to both parties. A two-of-three quorum signature is required for every commit.
- **Communication:** official coordination on the backbone; client instructions and price observations on local access or the direct service.
- **Margin:** one symmetric rule for both price directions, stated before either path is traced.

- **Extreme failure:** Earth gateway isolation for 72 h from $h = 230$ to $h = 302$, straddling the $h = 240.565$ future maturity. Mercury and Venus continue to serve non-Earth cross-cell work at a two-of-three quorum.
- **Blinded windows:** Venus is the designated bridge during recurring Earth–Mercury conjunction windows; the path-election rule selects it automatically because Earth and Mercury have blocked rays to the target.
- **Products:** Ares spot, NeoDollar transfer, fully escrowed fixed loan, capped cash-settled future, capped option, simple bond, fully backed settlement unit.

3 Architecture

3.1 Anchor selection

Table 1 reports the median backbone packet loss from each candidate anchor to the other eight settlements over the 200-year scan, computed from the supplied loss law $p = 1 - e^{-0.02d}$ with exact retarded distances. Earth, Mercury, and Venus are the three lowest-loss anchors and form the best inner-system triangle.

Table 1: Backbone loss from candidate anchors, 200-year scan. Lower is better. Weighted score double-counts the Neptune column.

Candidate	Avg. loss to 8 (%)	Loss to Neptune (%)	Weighted score
Mercury	15.86	45.79	1.0744
Venus	16.20	46.04	1.0828
Earth	16.60	46.39	1.0938
Mars	17.07	46.86	1.1140
Ceres	18.72	48.15	1.1715
Jupiter	22.05	50.62	1.2379
Saturn	27.73	54.71	1.3733
Uranus	39.18	62.66	1.6450
Neptune	50.15	0.00	0.0000

3.2 Path-elected initiator

Every cross-cell workflow begins with local locks. The originating cell places a durable hold on its own asset, then computes the best-placed anchor for the transaction using current retarded geometry. The choice minimises estimated total commit latency

$$T(A) = d(S_1, A) + 2 \min_{B \neq A} d(A, B) + d(A, S_2),$$

where S_1 and S_2 are the two participating settlements and d is the one-way backbone route delay at the moment of proposal. Ties break by lowest node ID (Mercury 1, Venus 2, Earth 3). The originating cell sends its signed instruction to that anchor, which becomes the initiator for this transaction.

The initiator sends PREPARE to the other two anchors. Each receiving anchor checks the transaction ID for uniqueness and its local view for consistency, then replies COMMIT or ABORT. If the initiator receives at least one COMMIT (giving a two-of-three quorum including itself), it signs the commit and broadcasts COMMIT to all affected cells. Each cell applies the commit in sequence; COMPLETE is reported only once every affected site exposes its new spendable balance.

No global monotone sequence number is required. Each anchor keeps its own journal with its own local ordering. The authoritative record is the union of the three anchor journals: a transaction is final when at least two anchors have signed its commit. Because every transaction ID is unique and every cell locks assets before proposing, no two final transactions touch the same asset. This is what preserves no-double-spend without a single writer.

Worked example. For an Earth–Venus trade, $T(\text{Earth}) = 2.028 \text{ h}$, $T(\text{Venus}) = 2.308 \text{ h}$, $T(\text{Mercury}) = 3.215 \text{ h}$; Earth is elected. For a Mercury–Venus trade with the same geometry, $T(\text{Mercury}) = 0.086 \text{ h}$, $T(\text{Venus}) = 0.086 \text{ h}$, $T(\text{Earth}) = 1.380 \text{ h}$; Mercury and Venus tie and Mercury wins on lowest node ID, saving about 1.3 h over an Earth-only rule. In all five S1 baseline scenarios in the appendix, Earth is a party, so Earth is elected every time; the rotating rule matters most for inner-system pairs.

The state machine. Every transaction moves through:

State	Transition rule	Financial effect
DRAFT	Client signs at home cell; cell checks identity, free balance, version, product limit.	None.
LOCKED	Each required cell places a durable hold and signs a lock.	Asset unavailable for other orders.
ELECTED	Originating cell selects the initiator by $T(A)$ and sends the signed instruction.	No debit or credit yet.
READY	Initiator has every lock plus at least one peer COMMIT; quorum of two reached.	No debit or credit yet.
COMMITTED	Quorum signs one balanced record; initiator broadcasts COMMIT.	Authority changes once. Terminal.
APPLIED	Each cell applies the commit in sequence.	Remote spendability follows signed local apply.
CANCELLED	Quorum issues terminal cancel only if never committed.	Locks release only on this certificate.

3.3 Quorum and failover

The quorum is any two of the three anchors. No leader election is required: every transaction carries its own initiator, and the quorum is formed per transaction. If one anchor is isolated, the other two continue to form quorums for new work. If two anchors are isolated, no new cross-cell commits can be signed; existing locks and records are preserved.

When Earth is isolated, Mercury and Venus form the quorum. They can commit any transaction whose elected initiator is Mercury or Venus. Earth-side cells cannot originate new cross-cell transactions during isolation, but their existing locks remain durable. When Earth returns, it syncs from the Mercury and Venus journals and resumes as a fully available initiator.

Because every transaction has a unique ID and every cell locks assets before proposing, the anchors do not need a shared monotone sequence. They only need to agree, per transaction, that the asset view was consistent at proposal time. This is what makes the rotating initiator safe: the safety property lives in the cell’s lock and the quorum signature, not in the anchor’s ordering.

3.4 Blinded-window bridging

Every 115.88 days Earth and Mercury align on the same side of the Sun. If an outer planet is near opposition from Earth, both Earth–outer and Mercury–outer straight-line paths can cross the 0.10 AU solar exclusion sphere. The Triad calls this a *blinded window*. Venus, at quadrature, is the natural bridge: it can see both the inner pair and the outer planet. The path-election rule selects Venus automatically in a blinded window, because Earth and Mercury have blocked rays to the target and therefore infinite estimated latency, so $T(\text{Venus})$ is the only finite value. Venus initiates the two-phase commit and the second quorum vote comes via Relay A or B from whichever of Earth or Mercury can see the relay. Table 2 gives a worked example.

Table 2: Blinded-window example at $t = 1917.79$ days. Longitudes in degrees.

Body	Longitude	To Neptune	Blocked cone	Status
Earth	87.65	180.04	5.7°	Blocked
Mercury	102.80	164.89	15.0°	Blocked
Venus	242.60	25.09	8.0°	Clear

3.5 Institutions

The Triad charts twelve institutions: three anchor replicas (Earth, Mercury, Venus) and nine other institutions. Each has its own direct quota of 12 packets per rolling 24 h; backbone originations count against the shared 600. Institutions start with nothing and are funded only by fees and approved capital contributions after hour 0.

Table 3: Chartered institutions.

Institution	Settlement	Role
Earth Anchor Registry (CCR)	Earth	Anchor replica; journal
Mercury Settlement Trust	Mercury	Anchor replica; journal
Venus Bridge Authority	Venus	Anchor replica; journal; blinded-window bridge
Backbone Operations Authority	Earth	Session scheduling; route monitoring; quota accounting
Price Bureau (PBS)	Earth	Signed price observations
Default Resolution Board	Earth	Administers the default waterfall
Arbitration Panel	Earth	Resolves contradictory records
Deep Space Guarantor	Jupiter	Posts guarantee capital for outer-planet clients
Neptune Settlement Agent	Neptune	Local access and client onboarding
Communications Coordinator	Mercury	Assigns pinned routes; tracks timers
Relay Health Monitor	Venus	Publishes relay status on the direct service
Outer-System Gateway Agent	Saturn	Local access for the outer four settlements

3.6 Authority and the authoritative record

The authoritative record is the union of the three anchor journals: a transaction is final when at least two anchors have signed its commit. Each anchor publishes its journal; clients and cells query any anchor for the state of a transaction ID. Because every transaction ID is unique and every cell locks assets before proposing, the union is consistent: no two final transactions touch the same asset at the same time.

A local cell’s signed lock is the enforceable barrier against a second spend. A stale balance display is never spend authority. A packet receipt is not a financial record. A COMMIT signed by a two-of-three quorum is the only authoritative financial event. The “one serial writer” intuition is preserved as one final record per transaction ID, not one writer per epoch.

3.7 Communication topology

Every anchor has a direct backbone link to Relay A and Relay B. Client settlements reach the anchors by backbone for official coordination or by direct service for client communication. Typical pinned routes are:

- Mercury $\rightarrow$ Relay A $\rightarrow$ Venus (two hops).
- Earth $\rightarrow$ Relay A $\rightarrow$ Venus (two hops).
- Earth $\rightarrow$ Relay B $\rightarrow$ Mercury (two hops).
- Neptune $\rightarrow$ Relay A $\rightarrow$ Earth (two hops); Neptune $\rightarrow$ Relay B $\rightarrow$ Venus is the alternate.
- Mars $\rightarrow$ Relay A $\rightarrow$ Earth (two hops).
- Ceres $\rightarrow$ Relay A $\rightarrow$ Earth (two hops).

- Jupiter $\rightarrow$ Relay B $\rightarrow$ Earth (two hops).
- Saturn $\rightarrow$ Relay A $\rightarrow$ Earth (two hops).
- Uranus $\rightarrow$ Relay A $\rightarrow$ Earth (two hops).

Relay A to Relay B is a legal third hop when geometry favours it, but the Triad pins routes without it for the baseline because a two-hop route has fewer failure points.

3.8 Traffic classification

Every message is classified by what it does, not by its label. Official coordination — anything that creates a shared exchange record or settlement authority — uses the backbone. Client orders, instructions, and observations use local access or the direct service. A signed price observation is client communication; a margin call or settlement instruction is official coordination and uses the backbone.

4 Operating Rulebook

4.1 Session lifecycle

A session is a pinned simple route (no repeated node, at most three backbone links) and its reverse, established by SYN, SYN-ACK, and ACK. The initiator may send data after receiving SYN-ACK and queuing the final ACK. The receiver releases data to the application only after the final ACK. A session may be reused; a replacement after expiry or reset is a new session and costs one originated packet. Sessions expire after seven days without a received session packet, or on a declared reset. Session identities are never reused.

4.2 Transport timers

SYN, SYN-ACK, and each data packet have at most four endpoint attempts. The hop retry timer is $R_h = 2 \times$ one-way flight +60 min, at most four launches per hop. The endpoint retry timer is $R_e = 2T_0 + 24$ h, where T_0 is the predicted empty-queue one-way route delay at enqueue; at most four attempts. Automatic retries and transport acknowledgments are quota-exempt but count in communication totals.

4.3 Order submission and matching

A client submits to its local gateway agent. The agent forwards to the elected initiator over direct service or, if co-located, by local access. The anchor’s matching engine matches by price–time priority: better price first, earlier submission next, ascending client node ID, ascending packet sequence number. A match is provisional until clearing confirms it.

4.4 Clearing: two-phase commit with path-elected initiator

Clearing uses two-phase commit over the backbone:

1. **Elect.** The originating cell computes $T(A)$ for $A \in \{\text{Earth, Mercury, Venus}\}$ using the current retarded route delays, breaks ties by lowest node ID, and sends its signed instruction to the elected anchor.
2. **Prepare.** The initiator sends PREPARE with trade details, proposed encumbrances, a transaction ID, and its local journal version to the other two anchors.
3. **Vote.** Each receiving anchor checks encumbrances on its replica and replies COMMIT or ABORT.
4. **Commit.** If the initiator plus at least one peer have voted COMMIT, the quorum signs and the initiator broadcasts COMMIT. Otherwise the initiator broadcasts ABORT and all locks release.

The protocol is idempotent: a duplicate PREPARE with the same transaction ID is acknowledged but not re-executed. In a blinded window Venus is elected automatically, and the second vote comes via a relay from whichever of Earth or Mercury can see it.

4.5 Settlement moments

Settlement has three reportable moments:

- **Discharge:** the total amount owed is extinguished on the authoritative record (at least two anchor journals).
- **Backed claim:** the winner holds a recorded claim backed by assets already encumbered.
- **Spendable:** the winner can use the amount as a balance at the winner’s own settlement; the loser’s encumbrance is debited or released to match.

For a value move, the spendable moment is the later of the two parties’ local updates. The Triad reports all three moments separately.

4.6 Margin rule

One rule applies to both price directions and uses only contract terms, received observations, positions, and declared risk assumptions. For N contracts of multiplier M at opening price P_0 :

$$\text{notional} = N M P_0, \quad \text{payoff} = N M \text{clip}(P_1 - P_0, -0.30P_0, +0.30P_0).$$

Each side posts $N M 0.30 P_0$ in cash. Initial margin for short equity is $0.50\times$ position value; for short options, $0.25\times$ notional. Maintenance margin is $0.75\times$ initial. For a capped future or option whose maximum possible loss is less than or equal to the posted initial margin, the position is fully collateralized and no maintenance call is generated; the maintenance rule applies only to residual uncapped or interim risk not covered by the cap. This is the rule used in the appendix Scenario C/D, where the $\pm 30\%$ cap equals the 30% initial margin. For uncapped or residual risk, a margin call fires when equity falls below maintenance; forced liquidation fires when equity falls below $0.50\times$ maintenance after a 24 h cure period. The rule is symmetric: long and short positions of the same notional post the same initial margin.

4.7 Default waterfall

If forced liquidation does not cover the obligation, the Default Resolution Board applies the following waterfall in order:

1. The defaulting client’s remaining encumbered assets at the clearing location.
2. The Deep Space Guarantor’s posted guarantee capital, up to its stated limit.
3. The Triad Clearing Council’s default fund, funded only from fees and approved contributions after hour 0.

No new assets are created. A funded default is reported with the backed-claim moment and the spendable moment, exactly as for any settlement.

4.8 Contradictory records

If two anchor journals disagree, the Arbitration Panel at Earth resolves using transaction IDs and journal versions. The two journals that agree win, because two of three is a quorum. If all three disagree, the Panel reconstructs state from durable records and issues a binding ruling on the backbone.

5 Products and Risk

5.1 Products

Table 4: Supported products.

Product	Terms	End of life
Equity (Ares Habitat)	Cash-versus-shares spot transfer in one quorum commit. Buyer may use shares locally only after apply.	Holder keeps the share; no maturity.
NeoDollar transfer	Serial two-phase transfer. Sender’s cash locked through commit; receiver spends after local apply.	Immediate on apply.
Fixed loan	Lender locks principal; borrower escrows full dated repayment before draw. Due at signed 240 h term.	Repayment at maturity; no automatic write-off.
Cash-settled future	One maturity print, capped $\pm 30\%$ from opening; 30% notional margin each side.	Final payment at maturity; interim marks are not separate obligations.
Capped option	European; cap at 150% of strike; premium paid up front; 25% notional margin for writer.	Final payout at expiry, capped.
Simple bond	500-hour term; fixed coupon (appendix Scenario B uses 5% and 700 pledged Ares shares); principal at maturity; issuer’s encumbered assets.	Principal and coupon at maturity.
Settlement unit (SU)	Issued by the Venus Bridge Authority; fully backed by encumbered NeoDollars at Venus.	Redeemable at par.

5.2 Service table

Table 5: Service table. Y = offered; R = offered with restriction; — = not offered.

Product	Mer.	Ven.	Ear.	Mar.	Cer.	Jup.	Sat.	Ura.	Nep.
Equity	Y	Y	Y	Y	Y	Y	Y	Y	Y
Future	Y	Y	Y	R	R	R	R	R	R
Option	Y	Y	Y	R	R	R	R	R	R
Bond	Y	Y	Y	Y	Y	Y	Y	Y	Y
SU	Y	Y	Y	Y	Y	Y	Y	Y	Y

Restriction R means the client at a non-anchor settlement posts an additional 5% margin in NeoDollars at the clearing location, because the local price observation is released less often. Anchors do not carry the restriction.

5.3 Price observations

The Price Bureau at Earth releases each signed observation as a single packet. The observation is distributed on the direct service under PBS’s own quota; in the appendix traces, the direct advisory for a futures maturity is labeled by its transport leg (e.g., V→E for the packet reaching Earth). The named observation for a future is the final price at maturity. The named observation for a bond is the coupon rate, fixed at issue. The named observation for an option is the expiry price. No direct bulletin authorises settlement; the quorum’s local signed source and backbone outcome remain authoritative.

6 Worked Spot Flow

Amina at Earth buys 100 Ares Habitat shares from Bayo at Mars for \$10,000. Each cell locks its own leg at $h = 0$. The originating cell elects the initiator by $T(A)$; with Earth and Mars as the two parties, Earth is elected. Bayo’s Mars cell sends its signed lock certificate via Mars–Relay A–Earth. The Earth anchor sends PREPARE to Mercury and Venus; both reply COMMIT; once Mars’s lock certificate arrives, Earth signs and broadcasts. The Earth–Relay A–Mars commit makes Bayo’s proceeds spendable at Mars.

Table 6: Spot flow, conditional no-loss trace.

h	Actor / knowledge	Action	Packet	State after
0	Amina knows her cash; Bayo knows his shares.	Earth locks \$10k; Mars locks 100 shares; Earth elected initiator.	Local.	No trade yet; both legs unavailable.
0.0006	Earth anchor has Earth’s lock.	PREPARE to Mercury and Venus.	Backbone E–A–Mer, E–A–Ven.	Pending.
0.0500	Mercury anchor receives PREPARE.	COMMIT.	Backbone Mer–A–E.	Replica updated.
0.0800	Venus anchor receives PREPARE.	COMMIT.	Backbone Ven–A–E.	Replica updated.
0.1300	Earth anchor has both votes.	Quorum reached; awaiting Mars lock certificate.	Backbone Mer–A–E, Ven–A–E.	No commit yet.
0.5651	Mars lock certificate reaches Earth.	Initiator has every lock; signs and broadcasts COMMIT to Mercury, Venus, and Mars.	Backbone E–A–Mer, E–A–Ven, E–A–Mars.	Trade final at Earth.
1.1302	Mars receives signed COMMIT.	Mars applies once.	Backbone E–A–Mars.	Bayo’s \$10k spendable; trade complete at both sites.
1.6953	Earth receives Mars APPLIED.	Earth closes workflow.	Backbone Mars–A–Earth.	No further balance change.

Why the order cannot run twice. The Mars cell removes the 100 shares from free inventory before sending its instruction. If a packet disappears, the lock remains. If a COMMIT was signed by a quorum, no anchor will issue CANCEL for that ID. A transport ACK changes no financial state.

7 Service at All Nine Settlements

All live cross-cell products require a two-of-three anchor quorum. Each row gives the selected one-way backbone route from the named settlement to the nearest anchor at $h = 0$ and $h = 300$; availability is the fraction of the following 24 h in which every link on the fixed route can launch, sampled every 5 min using exact Kepler geometry.

Table 7: Access table at $h = 0$ and $h = 300$ to the nearest anchor. All selected routes sample at 100% at both snapshots.

Home cell	$h = 0$ route; min	$h = 300$ route; min	Restriction
Mercury	Local (anchor); 0.017	Local (anchor); 0.017	None
Venus	Local (anchor); 0.017	Local (anchor); 0.017	None
Earth	Local (anchor); 0.017	Local (anchor); 0.017	None
Mars	Mars–A–Earth; 33.907	Mars–A–Earth; 33.439	5% extra margin on futures/options
Ceres	Ceres–A–Earth; 33.370	Ceres–A–Earth; 32.244	5% extra margin on futures/options
Jupiter	Jupiter–B–Earth; 50.923	Jupiter–B–Earth; 50.229	5% extra margin on futures/options
Saturn	Saturn–A–Earth; 79.419	Saturn–A–Earth; 78.862	5% extra margin on futures/options
Uranus	Uranus–A–Earth; 158.255	Uranus–A–Earth; 156.719	5% extra margin on futures/options
Neptune	Neptune–A–Earth; 250.480	Neptune–A–Earth; 250.052	5% extra margin on futures/options

At $h = 0$ the worst first-hop to an anchor is Neptune at 250.480 min one way. A completed Neptune-to-anchor-to-Neptune exchange can occupy about 8.35 h (the appendix S3.3 two-way completion rule) even with no random losses.

8 Network Promises

Every official inter-cell message uses the backbone. Each route is simple, at most three links, pinned for the session, and recomputes the moving receiver’s light time at every launch. A relay waits one second before its next link. Link receipts, endpoint handshakes, acknowledgments, retries, and application data all count as traffic.

Table 8: Service promises.

Service	Rules applied	If it does not arrive
Backbone	600 originated packets / rolling 24 h systemwide; 1 packet/s/directed link, FIFO queue 10,000. Hop timer $R_h = 2 \times \text{flight} + 60$ min, up to 4 launches; endpoint timer $R_e = 2T_0 + 24$ h, up to 4 attempts; packet TTL 30 d.	Keep the lock and same transaction ID. Query any anchor after restore; expired packet does not erase an obligation.
Direct client bulletin	12 originated packets / principal / rolling 24 h; at least 60 s apart; no receipt or retry. PBS uses one-copy advisory messages.	Client may miss the observation. The quorum’s local signed source and backbone settlement remain authoritative.
Local access	1 s each way, lossless, no quota.	Local product continues if it needs no remote price, record, or balance.

Admission. The quorum admits at most 40 new cross-cell workflows per rolling 24 h, and only when the shared originated-packet budget has fewer than 480 new packets in that window. Excess work is rejected before a new lock is taken; already-locked work is never silently discarded. A no-loss time is conditional, not a delivery promise. Random loss gives no finite unconditional deadline.

9 Stress Cases and Ranked Risks

9.1 Chosen incident: Earth gateway isolation

The stress run isolates the Earth gateway for 72 h from $h = 230$ to $h = 302$, exactly across the $h = 240.565$ future maturity. This is a damaging allowed incident for the Triad because Earth is the destination of a large share of demand and because direct Earth client packets are lost. During isolation, Earth-side cells

cannot originate new cross-cell transactions and Earth cannot serve as an initiator, but Mercury and Venus continue to serve cross-cell work between non-Earth parties at a two-of-three quorum. Existing Earth-side locks remain durable.

Table 9: Earth gateway isolation, $h = 230$ to $h = 302$.

Time	Event / what each side knows	Financial state
0–230	Future open; 6-day keeplive at $h = 144$ arrives.	\$75k margin each side locked locally.
230	Earth gateway isolation begins.	No new cross-cell fills through Earth; Mercury–Venus quorum takes over non-Earth work.
230.1	Mercury + Venus detect Earth loss; quorum confirms active.	New commits signed by two of {Mercury, Venus}.
240.565	PBS final $P = \$130$ at Earth; direct advisory to Mars is lost. Mercury and Venus sign the final outcome at quorum.	Amina’s local Earth margin handled at Earth; Bayo’s \$75k margin stays locked at Mars until quorum commit arrives.
265.7, 290.8	$R_e = 2T_0 + 24$ h; two retries fall inside isolation.	Same quorum; no duplicate payoff, no release of Mars hold.
302	Isolation ends; routes via Relay A are clear.	Fresh workflows resume; Earth syncs from Mercury/Venus.
303.1	First new two-phase spot completion under full three-anchor quorum.	Full service restored.
316.5	Old direct-email to Earth’s primary register is rejected as an outdated view; the quorum commit had already applied.	Original liability discharged once.

9.2 Ranked risks

Table 10: Ranked risks.

Rank	Failure	What breaks	Treatment and residual cost
1	Earth gateway isolation	Earth-as-initiator unavailable; direct Earth packets lost.	Mercury + Venus continue as a quorum for non-Earth work; Earth-side origination and price release resume after resync.
2	Common solar closure	A named route may disappear despite low per-launch loss.	Use a second allowed relay when clear; otherwise pause. No multiplying independent-copy probabilities across a common closure.
3	Queue / quota exhaustion	FIFO delay can invalidate naive timing; full queue drops newest.	40-workflow admission gate; no new cross-cell acceptance if a budget or queue reservation is missing.
4	Stale or duplicate data	Late status could tempt a user or cell to spend an old balance.	Unique IDs, per-anchor journal versions, quorum-signed outcomes, durable locks, 60-day transport dedup.
5	Price-source outage	Signed observation delayed or key compromised.	One named PBS key and versioned source record; no final price until source/time is locally verified. Key compromise triggers suspension.
6	Long-term storage / funding	Unresolved claims and old history consume resources.	Admission cap, finite hot-ID retention, compact funded obligations, fee-funded archive expansion; refuse new work at limits.

10 Continuing Operation

The market has no planned end. T+30 days ends a packet, not its funded financial obligation. The anchors store each unresolved owner, amount, account site, lock, maturity, and terminal version until it is discharged. An old packet that expires is queried under a fresh session and ID; the original contract ID and hold remain. No session, quota, debt, or position is cleared by a test-window boundary.

- **Sessions** expire after seven days without a received session packet; open positions send a keepalive every six days. An outage longer than that requires a fresh three-way handshake; session IDs are never reused.
- **Packets** live at most 30 days; duplicate packet IDs are retained at least 30 days after the last receipt. A 128-bit monotonically allocated transaction counter plus principal namespace does not wrap at the admitted demand rate. Packet sequence numbers are 64-bit and never reset within a session.
- **Hot state** is bounded by 60-day retention and the fixed 38 directed links. The quorum caps concurrent open cross-cell obligations at 100,000; at that limit it rejects new work while continuing maturity and recovery. Final records compact to signed per-account deltas and a 512-byte terminal record; 40 completed inter-cell workflows/day need about 7.5 MB/year.
- **Archive** has a 1 TiB funded operating tranche. At 80% it must be extended from earned 1 bp fees or voluntary capital before new workflows are admitted. If not extended, new cross-cell contracts stop; balances, records, local service, and existing obligations continue.
- **Capital** comes only from fees and approved contributions after hour 0. Voluntary subscriptions may expand operating capital; no fee or subscription creates money.
- **Demand rate** accepted: at most 50 financial workflows total (no more than 40 cross-cell) and 5,000 direct client packets per rolling 24 h systemwide. Above that rate, users get a signed queue/rejection result before any asset lock.

11 Deployment Assessment

The baseline folds rotation, ground visibility, satellite handoff, and planetary shadow into the 1 s lossless local access. Two external effects are worth planning for.

Table 11: Deployment effects.

Effect	Numeric estimate and source	Effect on guarantees
Solar plasma at Mars conjunction	NASA describes Earth–Mars communication quiet periods lasting a few weeks; one NASA account plans reduced operations for about two weeks. Use 14-day Mars gateway communications moratorium per conjunction as a planning overlay.	Do not accept a new Mars cross-cell fill inside the overlay. Existing shares, cash, and margin remain locked; local books continue. The 0.10 AU ray test alone is not a radio-quality guarantee.
Shared deep-space contact capacity	NASA OIG reports DSN demand has exceeded supply by as much as 40%. In a normalized example, capacity 100 units faces 140 requested; 40 units (28.6% of requests) unmet.	Treat all non-guaranteed queue delay as unbounded. The quorum reserves only admitted work; if contact allocation falls below the fixed-model assumption, reject or suspend new cross-cell flows without freeing old collateral.

These figures are planning analogies from present-day Mars and DSN operations, not measurements of the fictional network. The scan uses the challenge’s frozen Kepler model; it is not a 200-year forecast of the real solar system.

12 Promises, Costs, and Limits

Guaranteed by the rulebook.

- No cash or share is available for two uses at once: its home cell records one durable lock before a remote prepare.
- No shared record changes without a COMMIT signed by at least two of the three anchors. Each transaction ID has one terminal outcome.
- A final future payment is covered by posted cash under the capped payoff; when the cap equals or is below the posted initial margin, no maintenance call is generated (appendix Scenario C/D). A loan repayment is fully escrowed before principal is drawn. A short option's loss is capped at the strike cap.
- A transfer is reported complete only when every affected cell makes the new balance spendable. A backed claim may exist earlier and is labeled as such.
- No direct advisory copy, hop receipt, endpoint ACK, timeout, or packet expiry changes a financial balance.
- Under Earth isolation, Mercury and Venus continue to serve non-Earth cross-cell work at a two-of-three quorum; Earth-side origination and price release resume after resync.

Depends on connectivity or assumption.

- No unconditional cross-settlement delivery deadline. Random loss has no finite deadline; solar geometry, outages, queues, and quotas can delay work.
- The 24 h route percentages, the 200-year screen, and the 2000-year blinded-window overlay are sampled evidence, not proof of continuous future reachability.
- The model assumes unforgeable identities and honest operators. Key theft, collusion, or destruction of all replicas is outside the baseline and pauses service.
- A direct client may miss a price bulletin; the trade is not failed by that miss because the quorum's local source and backbone outcome remain authoritative.

What the participant pays. A 1 bp cash fee on each completed spot notional, loan draw, or futures opening notional funds the operator. Temporary collateral, travel-time illiquidity, and a 30-day packet lifetime are also costs. Non-anchor clients pay 5% extra margin on futures and options. The appendix reports peak encumbrance, asset-hours, capital ratio, packet counts, and fees instead of calling these costs "free reliability."

At a missing route, the model prefers a visibly pending, fully backed obligation to an apparently instant but unfunded settlement.